Kelan S. Zielinski

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Work & Leadership Experience

Motorola Solutions Inc. **Plantation, Fl**
Electrical Engineering Intern *May 2024 - August 2024*

- Successfully achieved optimal network impedance using a Smith chart by collaborating with RF engineers to analyze and adjust circuit parameters.
- Ensured signal integrity by implementing automated tests in LabVIEW to verify network matching, comparing RSSI values to predetermined benchmarks.
- Validated scattering (S2P) parameters on a Vector Network Analyzer (VNA) and redesigned the network on Advanced Design System (ADS) software.
- Performed root cause analysis using an oscilloscope to identify timing issues that resulted in unexpected power cycles between a radio and an uninterruptible power supply (UPS).

IoT-Driven Crop Production System **Coral Gables Fl**
Hardware Engineering Intern - University of Miami *January 2023 - May 2024*

- Designed and built an IoT-controlled hydroponics system to cultivate 2000+ plants per year within a footprint of less than 50 square feet.
- Developed a hardware prototype and validated components, including I2C and UART sensors and actuators, based on system requirements.
- Utilized an EDA simulation platform to model and optimize system circuitry for improved performance and reliability.

University Of Miami – Design For Testability Laboratory **Coral Gables, Fl**
Teaching Assistant *January 2024 - May 2024*

- Aided students in designing and implementing a digital signal processor in VHDL.
- Provided assistance with debugging timing issues in a large system.

HKN Honor Society (IEEE) – University of Miami Chapter **Coral Gables, Fl**
Eta Kappa Nu – Honor Society Member *February 2024 - Present*

- Supported students in mastering fundamental engineering concepts.
- Provided programming suggestions for HDL and C++ based projects.

Projects

FPGA-Based Quantum Emulator – VHDL, Python, ModelSim **May 2025**

- Designed and implemented a quantum system emulator on an FPGA using VHDL, modeling the time evolution of qubits based on their Hamiltonian.
- Optimized FPGA resource utilization for efficient matrix exponentiation and state vector updates to mimic quantum dynamics.

Advanced Digital Signal Processor – VHDL, ModelSim **April 2024**

- Implemented the ADSP-2100 digital signal processor to perform specific computations based on the instruction.
- Integrated Built-In-Self-Test (BIST) architecture to generate signatures and expose faults within the system design.

MIPS RISC CPU – VHDL, ModelSim, Quartus Prime **October 2024**

- Programmed, simulated, and implemented a 32-bit RISC processor.

Rotor-based Encryption Machine – Verilog, ModelSim, Quartus Prime **November 2023**

- Developed a Verilog-based model of the Enigma M3, featuring three dynamic rotors with rotational functionality and a reflector component.
- Verified data transfer and timing aspects of system design using RTL viewer and waveforms.

Technical Skills

Technical Skills: VHDL, Verilog, C++, C, Python, ARM Assembly, Soldering, PCB Design & Testing
Developer Tools: ModelSim, ADS, Cadence, PSpice, MATLAB, Intel Quartus Prime, KiCad, Unix, LabVIEW, AutoCAD
Test Equipment: Oscilloscopes, Spectrum Analyzers, Logic Analyzers, Network Analyzers, Multimeters, Signal Generators

Education

University of Miami **Coral Gables, Fl**
Bachelor of Science in Electrical Engineering – GPA: 3.72 *May 2025*

- **Relevant Coursework:** Digital System Design & Testing, Discrete Time Signals & Systems, Computer Organization & Design, Circuits Signals & Systems, Design for Testability(DFT) Laboratory, Electronics, Logic Design, Structured Digital Design, Electromagnetic Field Theory, Data Structures